Year 7 Physical Sciences

Workbook

Name _____

These activities are to be completed in the order they are in the booklet. Do not skip activities or teacher checks.

Activity 1

Graphing

Two students did an activity where they measured the amount of sugar that dissolves in 100 mL of water at different temperatures. Their results are recorded below.

Temperature of water (°C)	**Mass** of sugar dissolved (g)
5	10
10	15
20	18
35	20
45	30
50	35
60	38

Draw a line graph with **temperature** on the x-axis (horizontal), and **mass of sugar** on the y-axis (vertical). Don't forget to give a title to the graph.

Teacher check 

Activity 2

Open Science Quest 7 book to page 265 and read Activity 8.2.

Copy the table from the activity.

What to do	Observations	
	Changes in motion or shape	**What caused the change**

Get the equipment, conduct the activity and enter your observations into the table above.

Answer the three **Discuss and explain** questions in this activity.

1.

2.

3.

Teacher check 

Activity 3

The atom

All matter is made up of very small particles called atoms.

Atoms have a dense heavy nucleus in the centre.

This consists of two types of particles – protons which have a positive charge, and neutrons which have no charge.

Orbiting around the nucleus are negatively charged particles called electrons. The number of electrons is normally the same as the number of protons.

Protons have + charge

Neutrons have 0 charge

Electrons have − charge

Look at the diagram of the atom on page 273, Science Quest 7.

Now draw an atom of oxygen. This has 8 protons, 8 neutrons and 8 electrons.

Draw an atom of nitrogen. It has 7 protons, 8 neutrons and 7 electrons.

Now turn to page 276 and answer questions 1-10.

Teacher check 

Try Investigate, question 15 on page 276, Science Quest 7

What did you discover?

Activity 4

Turn to page 274 of Science Quest 7 and **read** Investigation 8.8:

The attraction of electricity

Draw a diagram of the investigation showing how the equipment will be set up.

Conduct the investigation.

Answer the Discuss and explain questions 1-5.

1.

2.

3.

4.

5.

Teacher check 

Activity 5

Research how a photocopier works.

Ask the teacher if you can access a computer to research how photocopiers work. Enter your findings below.

Activity 6

Read pages 277 and 278 of Science Quest 7.

Finish the following sentences.

Mass is

_____

_____It is measured in _____

Weight is

_____

_____ It is measured in

_____

Read Investigation 8.10 More than one force? On page 278 Science Quest 7.

Predict what will happen when you drop the coin and the paper at the same time.

_____

_____

Conduct the investigation.

Answer the Discuss and explain questions 1-6.

1.

2.

3.

4.

5.

6.

Teacher check 